

PIR - Sev 2 - Incorrect promo code badge appearing across PLP's, SRP's & PDP's for UK

M&S Digital&Tech *Post Incident Review*



Incident/ Problem Reference: INC000090678495

Incident Date: 21/03/2025

PIR Date: 27/03/2025

Author: Nilesh Singh

Attendees: Sumitro Chakravarty Karuppasamy, Ramachandran Natarajan, Srinivasan Hawthorn, Elaine
 ML Matthew, Leo Priyabrata Sahu SN Suneja, Nakul HS Hemal Shah Rajendran, Satish
 Nageswaran, Thulasitharen Dhanaseelan Mahalingam Kannan, Muthu Sundar, Shaxhy
 Chitluri, Kesava Manikanta

Technical Owner: Natarajan, Srinivasan Sreejith, Vineetha

Analyses the root cause and provide technical recommendations to prevent incident recurrence.

Develop and oversee the implementation of an action plan to address the root causes.

Work closely with cross-functional teams to gather information and insights.

PIR Reviewer: Webster, Charly

Validates the root cause analysis, corrective actions, and preventive measures before signing off, ensuring alignment with business and operational standards.

1. Incident Summary :

Provide a concise overview of the incident with brief update on issue, impact, root cause and resolution.

Internal shout was raised at 07:00 am on 21st Morning to promotion team stating Incorrect promo code badge appearing across high volume of PLP's, SRP's & PDP's, it is understood from the investigation the issues was due to manual error while promotion creation causing suboptimal customer shopping experience for UK (Web & App). Although these were system generated promo's (promocode) impact it was limited to Sub-optimal customer shopping experience as customers could start looking for a code that does not exist, but there was no direct commercial impact due to this incident.

To mitigate the issue, the promotion team identified the promo-id causing issue and it was immediately deactivated. As per process flow, promos were already presented to elastic search, indexed and cached it was required to rerun the nightly jobs to permanently fix. Meanwhile it was decided to go for quick tactical fixes to improve the customer shopping experience.

1. Optimizely experience - using web injection hide the offers/badges during page rendering process from the PLP's & PDP's.
2. Browse (onyx) side fix - It was observed post tactical fix, offers appear and disappear during the page rendering process which would eventually confuse customers and impact shopping experience. To overcome this rendering issue, the browse team was able to deploy a fix to

not show badges if it was for test promo code on the PLP's & SRP and applied filter to the promotions with test promo code to resolve issue on PDP's

3. Permanent Fix - To permanently fix the issue the WCS team proposed performing series of steps as recovery path for the resolution (Refer: - **Incident Resolution**)

2. Incident Chronology :

👉 Include a record of the data/time of key events. e.g. "07:36 Incident started, notified by ..." **keep fact based**".

07:30- Issue was highlighted through Internal shout, promotion team started Investigation and working for fix

08:15 - Promotion (Test_£10 off 210325 & Test_10% off 210325) causing the issue was deactivated as an emergency mitigation

08:35 - Promotion codes deactivated at WCS end.

09:08 - Product Assembly completed and Akamai cleared the cache.

09:45 - Still promotion is seen in the API response of the PLP's & PDP's

09:50 - Recovery options are being evaluated to identify risk vs mitigation as it is approx. 123k items seen impacted

10:00 - **Workaround** - Optimizely Experience (Web experimentation tool)

- Optimizely experience (hiding promo) by using Web injection to be evaluated to decide as an option for PLP & PDP pages (Web)

10:10 – Optimizely experience (Web injection) workaround tested successfully for one of the PLP's, (This is Tactical fix applicable only to customers accepting cookies).

10:15 - Tactical fix applied across all PLP's is live now.

10:35 - Optimizely experience (Web injection) workaround tested successfully for one of the PDP's.

- Tactical fix applied across all PDP is live now.

10:44 - **Permanent Fix**

WCS proposed below steps for permanent fix along with performance monitoring

- Removing the mappings at WCS DB
- Validate commerce API
- Product assembly validation and cache clear
- Elastic search full feed
- Search API cache clear
- PLP/SRP cache clear
- Remove strategic UI fix

11:00 - Flicker issue

- Flicker (PLP & PDP workaround) is being addressed by Browse team who are working on an amendment to correct that behavior.
- WCS testing the proposed permanent fix with one sample product.

11:37 - Browse side **tactical fix**

- Browse change to hide badges from PLP/SRP/PDP tested and looks good. Team will deploy the tactical fix from Browse (40 - 45 minutes approx. ETA)
- WCS Permanent fix - DBA has executed query to remove test promo from 123k products on WCS database. Teams are just now validating that the promo is removed from all products.
- Cache cleared for Product domain and Promotion/Price APIs

12:00 -

- Mobile Apps PDP issue is resolved. Cache clear initiated for Onyx PDP as the origin is now seeing the correct promotions.
- Search full feed for UK Clothing & Home has been initiated. (Rough ETA 1.5 hours)
- Cache cleared for Onyx PDP. PDP badges issue resolved for Web.

12:40 -

- Browse deploy to hide test promo badge is deployed to EUN & EUW. Origin for Browse is no longer showing the issue for PLP/SRP
- Stakeholders and dependencies informed that a cache clear will be performed on PLP for Browse. Initiated at 12:39.

- Optimizely tactical fix switched off for PDP, its currently fixed on all channels.
- Browse cache for SRP cleared.

13:00 - Permanent fix (**update**)

- Elastic search full feed in progress

13:35 - Service restored

- Elastic search full feed job completed at 13:29
- Search API cache clear completed and validated (Web)
- Validation looks good for App (Android & iOS)

24th Mar - Monday

- 17:23 - Onyx browse side tactical fix was reverted.

3. Incident Resolution

👉 *The actions taken to restore services and mitigate impact.*

Resolution followed couple of tactical fixes to control the issue till the series of steps at WCS were completed to fix it permanently. **(for understand the permanent fix (recovery journey) it is recommended to go through tech processing explanation in root cause section)**

- **Optimizely experience tactical fix** - using web injection method the offers were hidden during page rendering process. The workaround was tested and was pushed across PLP's & PDP's. It was observed that customers could see offers appearing and disappearing during the page loading process and could contribute towards poor customer browsing experience.
- **Browse tactical fix** - To overcome the rendering issue (badges appear/disappear), the browse team was able to deploy a fix to not show badges if it was for test promo code on the PLP's & SRP and applied filter to the promotions with test promo code to resolve issue on PDP's.
- **Permanent fix** - WCS team came up with series of steps to mitigate the issues from both Web and App.
 - Removing the mappings at WCS DB
 - Validate commerce API
 - Product assembly validation and cache clear
 - Elastic search full feed
 - Search API cache clear
 - PLP/SRP cache clear
 - Remove strategic UI fix

4. Root Cause

👉 *Refers to the fundamental reason (s) why an incident occurred including problem description, analysis techniques, contributing factors, evidence with concise root cause statements*

Cause

Based on investigation it is understood that a manual error while creating promotion in the system caused this issue, promotion team received a request for sample promo code for the purpose of testing, promo code was created by cloning the previously used template (manually) and accidentally missed to remove sparks promotion segment from the template (**no segment should be attached to this promotion**). Further promotion was not tagged which made it go live site wide for UK.

Tech Processing (explanation)

As there were no feature flag available to switch-off the promotion across impacted products, tech teams were onboarded to help with understanding processing flow and identifying recovery steps, usually for promocode based promotions, WCS will not generate mappings in XCATENTPROMO (table). But for Sparks promotions, WCS creates the mapping, and this mapping is exported in promotion API which further enables PLP/PDP/SRP to show the Sparks offer logos.

For the impacted promotion (Test_£10 off 210325), the Promotion team forgot to remove the customer segment '**Sparks Promotion Segment**' from the template, which made the script to calculate the XCATENTPROMO mappings and hence exported in promotion API.

Also, this promotion was created based on the private promocode (system generated promocode), "**loyaltyPromoCode**" tag was not sent in

promotion API response, so Elastic search considered it as a site wide normal promotion and displayed the promo badge across all the applicable products.

5. Findings

👉 *Refer to the key insights and conclusions drawn from the incident analysis - what went well/what went wrong highlighting gaps, failures, or inefficiencies.*

5.1. Observability –

👉 *Evaluating monitoring, logging, and alerting effectiveness in detecting the incident.*

The issue was highlighted via internal shout; there were no validation alerts in place to identify such issues.

Promotion and WCS teams worked towards configuring alerts for identifying unusual promotion combination as proactive measure before WCS jobs are scheduled to run.

5.2. Technical Challenges -

👉 *Identifying system, infrastructure, or architectural issues that complicated resolution.*

Challenge in identifying the impacted number of products across PLP's & PDP's, there is no available feature flag to switch off the badges across website and app. Reversion and recovery took longer time as it involves Elastic search feed and clearing cache across search API and PLP's/PDP's.

Mitigating the issue on App side took longer time as tactical fix applied on web was not applicable for app and couldn't be applied until permanent fix steps were completed.

5.3. MIM involvement & impact assessment -

👉 *Timely engagement of MIM and analysing the impact on business operations based on top 10 business processes*

Gap in impact assessment.

This wasn't a tech issue although recovery required tech support and hence the service lead was engaged, internal shout was received through one of the teams' channels, already the promotion team was informed and were working towards disabling the badges. There was delay in starting the bridge due to network issues

5.4. Incident handling/ stakeholder management –

👉 *Reviewing how teams collaborated to drive the incident to resolution and engaged with key stakeholders.*

The promotion team was already involved and was investigating to identify and disable the promo's causing issues. The tech ops team was informed with impact and business comms were sent to all the stakeholders.

The incident bridge was initiated bit late due to technical challenges (network issue)

5.5. Business Communication –

👉 *Clear, concise & timely communication to stakeholders, focusing on business impact and recovery actions.*

Email and text comms were shared with Sev-2 stakeholders, the delay in circulating and summarizing the communication highlights need to revisit the comms framework.

5.6. Supplier Engagement –

👉 *Assessing the responsiveness and support provided by external vendors and internal product teams.*

Mitigation Actions:

👉 *Agree actions for each of the findings to prevent recurrence, strengthen system reliability & stability, and improve response efficiency for future incidents*

	≡ Action	≡ Owner	📅 Due Date	≡ Remarks	🔖 Action Category
1	Product Assembly Cache Clear	 Chitluri, Kesava Manikanta	Fri, Mar 21, 2025	Completed	

2	Promotion API still showing the test promo. WCS team investigating	 Dhanaseelan Mahalingam	Fri, Mar 21, 2025	Completed	
3	Promotion is still active in WCS. Team will investigate and ensure promotion is removed + base cache clear.	 Dhanaseelan Mahalingam  Revathi Gopalakrishnan	Fri, Mar 21, 2025	Completed	
4	Promotion removed & Base Cache cleared, however promotion still showing up intermittently on API. WCS team reviewing whether nightly job is required to remove the promotion completely or whether there are any other options	 Dhanaseelan Mahalingam  Revathi Gopalakrishnan  Karuppasamy , Ramachandran	Fri, Mar 21, 2025	Completed	
5	2nd Product Assembly Cache Clear following promotion removal	 Chitluri, Kesava Manikanta	Fri, Mar 21, 2025	Completed	
6	WCS team checking if they can amend a product manually in the database	 Dhanaseelan Mahalingam	Fri, Mar 21, 2025	Completed	
7	.com Business Ops checking whether there are any priority PLPs need to be fastracked via a delta	 Khanna, Himanshu	Fri, Mar 21, 2025	Completed	

8	Mobile Apps team added to the bridge to see if a similar strategy to the Optimizely one used on the Onyx Web pages can be applied to iOS & Android App. Apps team confirmed they are unable to apply any Optimizely solution as a tactical fix to hide the badges.	Apps team /  Andreas Christophides	Fri, Mar 21, 2025	Completed	
9	Ops Hub team are working on masking badges via Optimizely web injection across PLPs and PDPs (started 10:05). PLPs were completed at 10:15. PDPs completed at 10:44. SRP completed at 11:30	Ops Hub	Fri, Mar 21, 2025	Completed	
10	The tactical fix has introduced a flicker effect on the pages it is hiding. Browse team are working on an amendment that will remove this flicker	SN Suneja, Nakul HS Hemal Shah	Fri, Mar 21, 2025	Completed	
11	Run Search Full Feed	PM Pattusalli Janakiraman, Maheshkumar	Fri, Mar 21, 2025	Completed	

12	Delta to run for top PLP's shared by ops hub in order to get the fix applied for them quicker. Being run in batches of 1k products.	PM Pattusalli Janakiraman, Maheshkumar	Fri, Mar 21, 2025	Completed	
13	Once Full Feed Completes Search caches should be cleared followed by PLP Cache Clears	Browse team + Product Domain	Fri, Mar 21, 2025	Completed	
14	Once promotions are removed from Promotion API, cache clears for PDP related path should be performed.	Search and Browse Team	Fri, Mar 21, 2025	Completed	
15	Revert PR prepared for Browse tactical fix. Action to revert this on Monday.	HS Hemal Shah SN Suneja, Nakul	Mon, Mar 24, 2025	Completed	

Actions as part of PIR

	≡ Action	👤 Owner	📅 Due Date	≡ Remark
1	Guidelines to create promotions fresh rather cloning pervious templates and SOP was updated with checklist and peer review guidelines.	👤 Natarajan, Srinivasan	Tue, Mar 25, 2025	Completed
2	Configured alert to detect improper combinations e.g. system coded combined with sparks segment.	👤 Dhanaseelan Mahalingam 👤 Revathi Gopalakrishnan	Tue, Mar 25, 2025	Completed

3	Business ops team to include checks for inappropriate badges while performing site walk (Sanity checks)	 Natarajan, Srinivasan	Tue, Mar 25, 2025	Completed
4	Exploring the feasibility of decoupling promotion displays from the main product assembly process	 Chitluri, Kesava Manikanta	Mon, Mar 31, 20...	We are not doing any development work in PA as its legacy API and consumers will use Federation to get similar functionality in future.

Please refer to [Major_Significant and Key Incident Tracker v1.0.xlsx](#) for progress updates on Mitigations

<Mandatory participants: Put it in the meeting invite. >

Technical Details and Useful Documents:

PX_PROMOTION_ID	NAME	STATUS	STARTDATE
727103884	Test_£10 off_210325-727103884	0	2025-03-21 00:00:00.001
727103885	Test_10% off_210325-727103885	0	2025-03-21 00:00:00.001

FitSizeBody FitColourBrandPrice

49 - 96 of 230 Items

Relevance



Test promo code

£35

M&S COLLECTION

Relaxed Tapered Fit Single Pleat Jeans

4.8★★★★★4 reviews



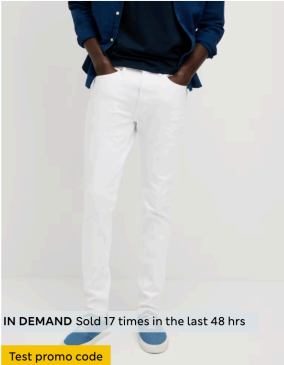
Test promo code

£35

M&S COLLECTION

Relaxed Tapered Fit Single Pleat Jeans

4.8★★★★★4 reviews



Test promo code

£45

M&S COLLECTION

Skinny Fit 360 Flex Stretch Jeans

4.1★★★★★9 reviews



Test promo code

£60

AUTOGRAPH

Straight Fit Japanese Selvedge Jeans

4.6★★★★★36 reviews